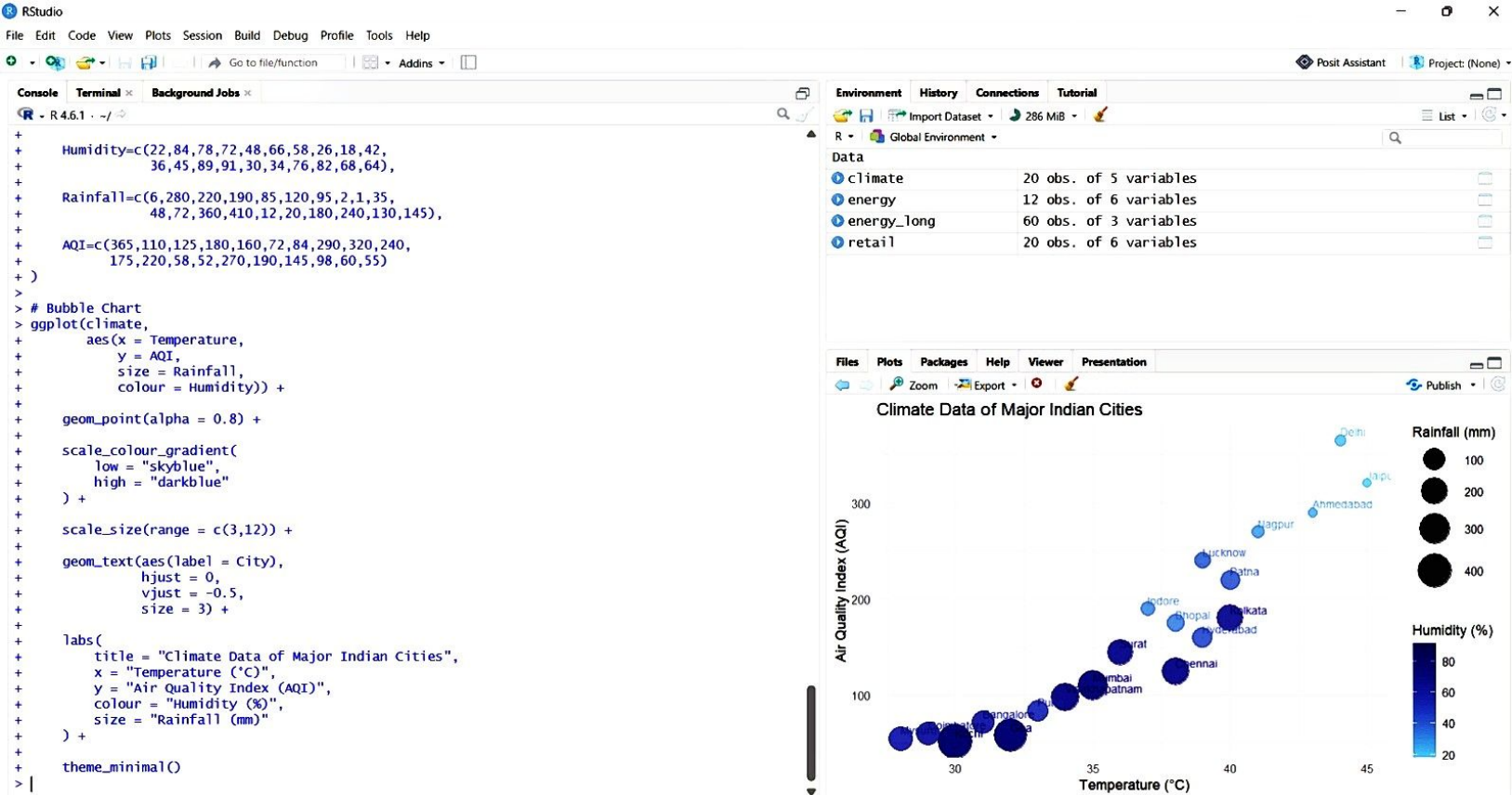




Console Terminal Background Jobs

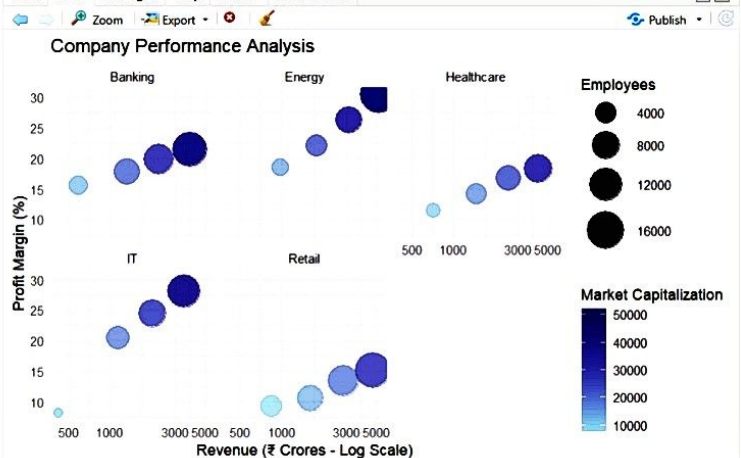
```
R - R 4.6.1 - ~/
+      24.5,19.8,16.7,13.4,26.2,
+      28.1,21.5,18.3,15.2,30.4),
+   Employees=c(1500,3200,2100,4200,2800,
+               5100,6800,3900,7200,4500,
+               8100,9500,6200,10300,7200,
+               11800,13200,8600,14500,16000),
+   MarketCap=c(8500,12000,9800,7600,14500,
+               18200,21500,16800,13200,24500,
+               28500,32500,24800,19200,38200,
+               42500,46800,35600,29800,52000)
+ )
>
> # Bubble Chart
> ggplot(companies,
+   aes(x = Revenue,
+       y = ProfitMargin,
+       size = Employees,
+       colour = MarketCap)) +
+
+   geom_point(alpha = 0.8) +
+
+   facet_wrap(~Sector) +
+
+   scale_x_log10() +
+
+   scale_size(range = c(3,12)) +
+
+   scale_colour_gradient(
+     low = "lightblue",
+     high = "darkblue"
+   ) +
+
+   labs(
+     title = "Company Performance Analysis",
+     x = "Revenue (₹ Crores - Log Scale)",
+     y = "Profit Margin (%)",
+     size = "Employees",
+     colour = "Market Capitalization"
+   ) +
+
+   theme_minimal()
>
```

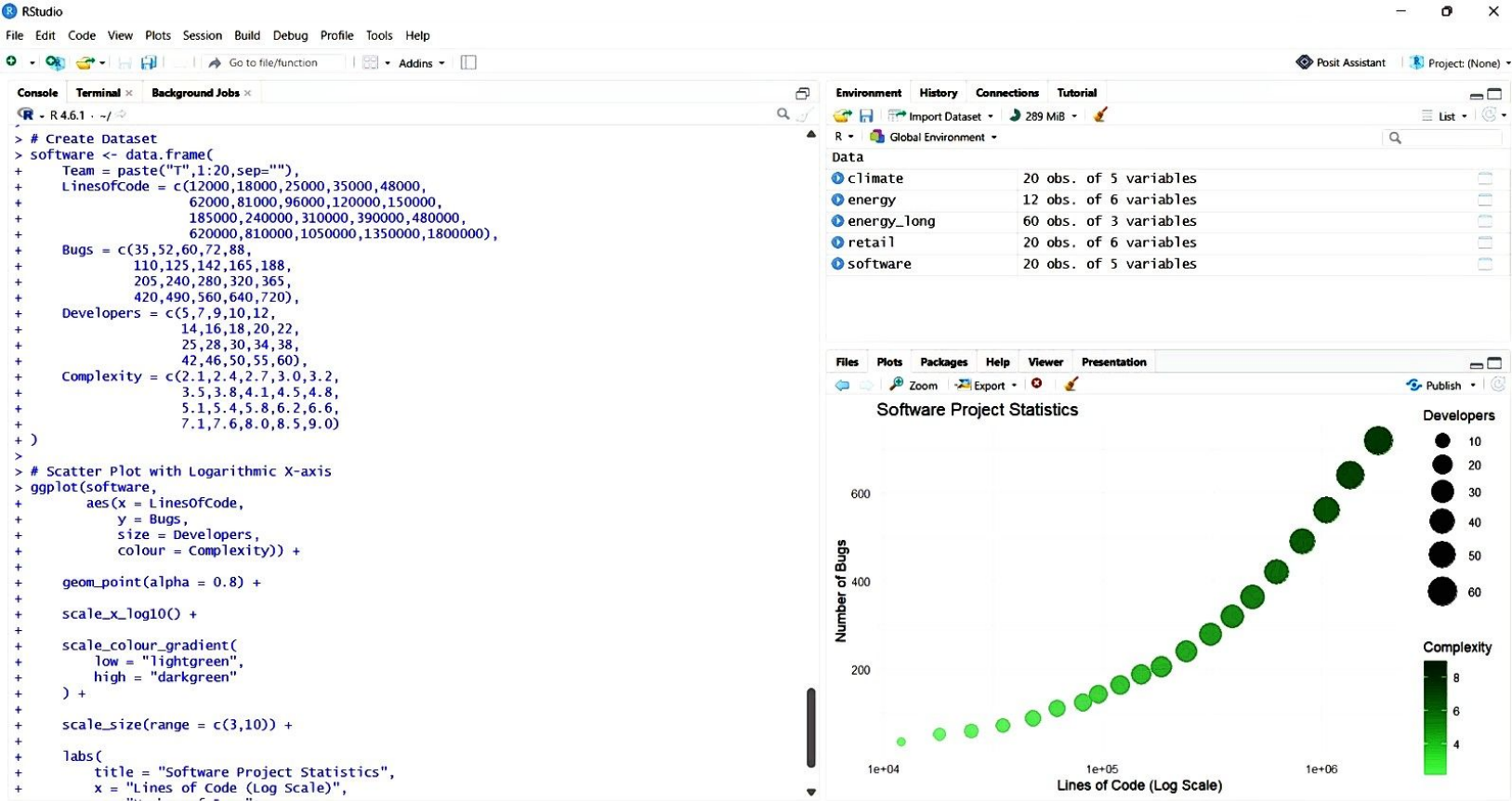
Environment History Connections Tutorial

R - Global Environment

Data		
climate	20 obs. of 5 variables	
companies	20 obs. of 6 variables	
energy	12 obs. of 6 variables	
energy_long	60 obs. of 3 variables	
retail	20 obs. of 6 variables	
software	20 obs. of 5 variables	

Files Plots Packages Help Viewer Presentation





Console Terminal Background Jobs

```
R > R4.6.1 ~ /
package 'ggplot2' successfully unpacked and MD5 sums checked
```

```
The downloaded binary packages are in
  C:\Users\HP\AppData\Local\Temp\Rtmp23T1p1a\downloaded_packages
```

```
> # Install ggplot2 (Run only once)
```

```
> # install.packages("ggplot2")
```

```
>
```

```
> # Load library
```

```
> library(ggplot2)
```

```
Need help? Try Stackoverflow: https://stackoverflow.com/tags/ggplot2
```

```
>
```

```
> # Create Dataset
```

```
> retail <- data.frame(
```

```
+   Branch=c("B1", "B2", "B3", "B4", "B5", "B6", "B7", "B8", "B9", "B10",
```

```
+           "B11", "B12", "B13", "B14", "B15", "B16", "B17", "B18", "B19", "B20"),
```

```
+   Region=c("North", "South", "East", "West", "North", "South", "East", "West",
```

```
+           "North", "South", "East", "West", "North", "South", "East", "West",
```

```
+           "North", "South", "East", "West"),
```

```
+   Category=c("Electronics", "Furniture", "Clothing", "Food", "Electronics",
```

```
+              "Furniture", "Clothing", "Food", "Electronics", "Furniture",
```

```
+              "Clothing", "Food", "Electronics", "Furniture", "Clothing",
```

```
+              "Food", "Electronics", "Furniture", "Clothing", "Food"),
```

```
+   Sales=c(450000, 620000, 180000, 920000, 350000, 720000, 250000, 1080000,
```

```
+           560000, 670000, 195000, 980000, 430000, 760000, 280000, 1150000,
```

```
+           510000, 690000, 220000, 1250000),
```

```
+   Profit=c(45000, 95000, -12000, 185000, 25000, 120000, 18000, 225000,
```

```
+            72000, 98000, -8000, 205000, 51000, 132000, 24000, 250000,
```

```
+            69000, 115000, 15000, 285000),
```

```
+   Customers=c(1200, 2500, 980, 4200, 1800, 3000, 1250, 5200,
```

```
+               2100, 3300, 1100, 4700, 1750, 3600, 1450, 5600,
```

```
+               2200, 3900, 1600, 6100)
```

```
+ )
```

```
>
```

```
> # Bubble Chart
```

```
> ggplot(retail,
```

```
+   aes(x = Sales,
```

```
+       y = Profit,
```

```
+       size = Customers,
```

```
+       colour = Category)) +
```

```
+   geom_point(alpha = 0.8) +
```

```
+ 
```

Environment History Connections Tutorial

```
Import Dataset 222 MiB
```

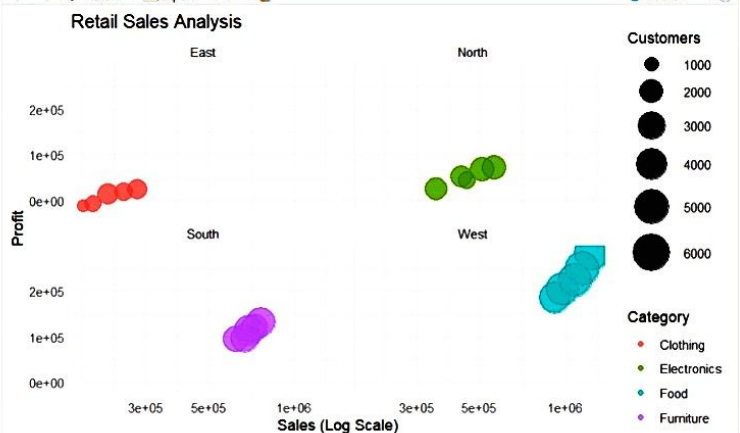
```
R Global Environment
```

Data

```
retail 20 obs. of 6 variables
```

Files Plots Packages Help Viewer Presentation

Zoom Export Publish



Console Terminal Background Jobs

```
R - R 4.6.1 ~ /
> library(ggplot2)
> library(reshape2)
>
> # Create Dataset
> energy <- data.frame(
+   Month = month.abb,
+   Coal = c(620,580,590,605,640,670,690,720,700,680,650,630),
+   Gas = c(310,295,305,320,340,355,360,370,365,350,335,320),
+   Solar = c(45,60,90,130,180,230,250,245,210,160,90,50),
+   Wind = c(80,95,120,150,170,185,195,190,175,150,110,90),
+   Hydro = c(140,150,160,170,180,195,205,210,200,185,165,150)
+ )
>
> # Convert data to long format
> energy_long <- melt(energy, id.vars="Month")
>
> # -----
> # Cartesian Coordinate System
> # -----
>
> ggplot(energy_long,
+   aes(x=Month,
+       y=value,
+       colour=variable,
+       group=variable)) +
+   geom_line(size=1.2) +
+   geom_point(size=2) +
+   labs(title="Monthly Electricity Generation",
+        x="Month",
+        y="Generation (Units)",
+        colour="Energy Source") +
+   theme_minimal()

Warning message:
Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
Please use 'linewidth' instead.
This warning is displayed once per session.
Call lifecycle::last_lifecycle_warnings() to see where this warning was generated.
>
> # -----
> # Polar Coordinate System
> # -----
```

Environment History Connections Tutorial

R Global Environment

Data

energy	12 obs. of 6 variables
energy_long	60 obs. of 3 variables
retail	20 obs. of 6 variables

Files Plots Packages Help Viewer Presentation

Zoom Export Publish

Monthly Electricity Generation (Polar Chart)

